

Juhyun Jung

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RESEARCH INTERESTS

Underwater Robotics, Bio-Inspired Robotics, Robotics, and Deep Learning

EDUCATION

Stanford University

M.S. in Mechanical Engineering

- GPA: 4.06/4.00; Advisor: Prof. Mark Cutkosky and Prof. Shuran Song

Stanford, CA

Sept. 2025–Present

Illinois Institute of Technology

B.S. in Mechanical Engineering, Minor in Artificial Intelligence

- GPA: 4.00/4.00, *Dean's List*

Chicago, IL

Aug. 2022–Dec. 2024

Ajou University

B.S. in Mechanical Engineering

- GPA: 4.47/4.50, *ranked 1st in class (1/170)*, *Dean's List*

Suwon, South Korea

Mar. 2018–Dec. 2024

RESEARCH EXPERIENCE

Biomimetics and Dextrous Manipulation Lab (BDML), Stanford University

Graduate Research Assistant · Advisor: Prof. Mark Cutkosky · Stanford, CA

Sept. 2025–Present

UMI-Underwater: Robust Manipulation (co-advised by Prof. Shuran Song)

Apr. 2026–Present

- Developing robust control policies for an underwater manipulator to achieve stable grasping under hydrodynamic disturbances.

Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots

[arXiv:2608.24724](https://arxiv.org/abs/2608.24724)

Sept. 2025–Apr. 2026

- Developed a novel tactile sensing modality inspired by animal whiskers, integrating fiber Bragg grating (FBG) optical strain sensors with compliant beams for underwater flow detection.
- Improved the signal-to-noise ratio by over 8× through experimental design and signal processing.
- Designed and built an embedded testbed with stepper-motor actuation and real-time data acquisition.

Computer Vision and Multimedia Laboratory (CVM Lab), Illinois Institute of Technology

Undergraduate Research Assistant · Advisor: Prof. Yan Yan · Chicago, IL

June 2023–Dec. 2024

- Developed an LSTM-based algorithm to classify mouse behavior from pose datasets.
- Built a Vision Transformer (ViT) classifier over full mouse video datasets, improving behavior prediction accuracy.
- Supported PhD students in evaluating multimodal models, including concatenation fusion and gated information fusion, for behavior classification.

Multiscale Bio-Inspired Technology Laboratory (MOST Lab), Ajou University

Undergraduate Researcher · Advisor: Prof. Je-Sung Koh · Suwon, South Korea

Jan. 2022–June 2022

- Designed and built a comprehensive experimental setup for dielectric elastomer actuator (DEA) research.
- Characterized DEA performance across frequency, electrode area, elastomer composition and thickness, voltage, and pre-strain, demonstrating a maximum strain of 12.3%.

PREPRINTS

1. H. Li, T. Tu, S. Yao, Z. Chang, **J. Jung**, X. Xie, L. Y. Chung, T.-A. Ren, G. Chen, and M. Cutkosky, "Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots", [arXiv:2608.24724](https://arxiv.org/abs/2608.24724), 2026.

PROJECTS	Understanding the Varied Economic Impact of Tax Increment Financing (TIF) Jan.–May 2024 Advisor: Prof. Robert Ellis · Consulting expert: Tom Tresser (Co-Founder, CivicLab) Analyzed the effect of TIF districts on median income in Chicago, showing a significant increase for the White population while other racial groups were largely unaffected, with the largest gains in White-dominant, high-income districts.
	Autonomous Mobility Competition — Vision-Based Lane Following Sept.–Dec. 2021 Designed a vision-based lane-following algorithm combining image processing and PID control to optimize real-time model-car performance; awarded 2nd place by HL Mando Corporation.
	System Innovation Competition · Advisor: Prof. Peom Park Mar.–May 2021 Analyzed limitations of VR technology such as cybersickness and designed solutions including accelerometer integration and sound effects, supported by market surveys; awarded 3rd place by KOSSE.
PRESENTATIONS	J. Jung, “Autonomous control using automated image analysis”, <i>Fall 2024 Armour R&D Expo</i>, Chicago, IL, Nov. 2024. (Poster) J. Jung, “Autonomous control using automated image analysis”, <i>Summer 2024 Armour R&D Research Immersion Exposition</i>, Chicago, IL, July 2024. (Poster) J. Jung, “Understanding the varied economic impact of Tax Increment Financing (TIF)”, <i>Ed Kaplan Family Institute Innovation Day</i>, Illinois Institute of Technology, Chicago, IL, Apr. 2024. (Poster) J. Jung, “Assessment of cybersickness in VR social media services”, <i>The Korean Society of Systems Engineering 2021 Conference</i>, Jeju, South Korea, June 2021. (Poster)
HONORS & AWARDS	Dean’s List , Illinois Institute of Technology 2022–2024 Dean’s List , Ajou University 2018, 2021, 2022 Academic Excellence Scholarship , Ajou University 2018, 2021, 2022 Department Chair’s Award for Academic Excellence , Ajou University 2019, 2022 2nd Place, Autonomous Mobility Competition , HL Mando Corporation 2021 3rd Place, System Innovation Competition , Korean Society of Systems Engineering 2021 Soldier of Valor Award , Republic of Korea Army 2020 Active Learner Self-Development Scholarship , Ajou University 2018
SKILLS & COURSEWORK	Programming & Tools: Python, C/C++, MATLAB, Julia, ROS 2, Linux (Ubuntu 22.04) Robotics: Robot Autonomy I (CS 237A), Collaborative Robotics (ME 326), Introduction to Robotics (CS 223A), Optimal and Learning-Based Control (AA 203) Machine Learning: Deep Learning (CS 230), Decision Making under Uncertainty (CS 238), Deep Reinforcement Learning (CS 224R)
ACTIVITIES	- Balanced Man Scholarship Chair, Sigma Phi Epsilon Fraternity (Illinois Beta) - Member, Tau Beta Pi Engineering Honor Society - Member, Climbing Club